



2 The line l_1 has equation $\mathbf{r} = \mathbf{i} + 3\mathbf{j} - \mathbf{k} + \lambda(\mathbf{i} - \mathbf{j} - 4\mathbf{k})$.

The plane Π contains l_1 and is parallel to the vector $2\mathbf{i} + 5\mathbf{j} - 4\mathbf{k}$.

(a) Find the equation of Π , giving your answer in the form $ax + by + cz = d$. [4]

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



(b) Find the acute angle between l_2 and Π .

[3]

[illegible]